SIC-POVMs, the Stark Conjecture, and Hilbert's Twelfth Problem: A Lean 4 Formalization via Paraconsistent Belnap Multilattices

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Abstract

We present a Lean 4 formalization establishing a three-level structural identification: *(i)* the Belnap multilattice axioms for Weyl–Heisenberg (WH) covariant symmetric informationally complete positive operator-valued measures (SIC-POVMs) in dimension $d = 2^n$; *(ii)* the Zauner conjecture for those dimensions; and *(iii)* the mixed-signature Stark conjecture for the ray class fields $K_d = \mathbb{Q}(\sqrt{(d-3)(d+1)})$ — a special case of Hilbert's Twelfth Problem for real quadratic fields. The core equivalence `hilbert_embedding_equiv_zauner` is proved by `rfl`: the Hilbert-space embedding of the Belnap multilattice into $\mathbb{C}^{2^n}$ and the Zauner conjecture at $d = 2^n$ are definitionally identical propositions. The Belnap structural skeleton—orbit cardinality 4^n, Frobenius closure $\mu \circ \delta = \mathrm{id}$, join-equiangularity, and a structural Born rule—is proved unconditionally with zero `sorry`s. The remaining open content is precisely axiomatized: three named axioms carry the exact arithmetic geometry of Stark units acting on WH frames. A proof of the mixed-signature Stark conjecture closes all three levels simultaneously. For dimension $d = 12$ we close the existence question outright. We construct an exact fiducial whose twelve coordinates live in an explicit 2048-dimensional commutative $\mathbb{Q}$-algebra with involution, machine-check the norm and all 143 Weyl–Heisenberg overlap identities there by exact rational computation, and transfer them to $\mathbb{C}^{12}$ along an explicit ring homomorphism. The resulting theorem `crystal_forces_d12_sic` is sorry-free and rests on no axiom beyond Lean 4's standard foundations and the `native_decide` compiler trust; the existence axiom it replaces has been deleted from the development. To our knowledge this is the first machine-checked existence proof of a SIC-POVM in any dimension.

Contents

*Thanks are owed to Harry T. Larson; cf. [4].

1 Introduction

Symmetric informationally complete positive operator-valued measures (SIC-POVMs) are minimal tomographically complete quantum measurements: d^2 rank-1 projectors in $\mathbb{C}^d$ with constant pairwise Hilbert–Schmidt inner products. Their existence in every finite dimension is widely conjectured (Zauner's conjecture, 1999 [8]) and verified numerically through $d = 2406$, yet no general proof exists.

Appleby, Bengtsson, Flammia, and collaborators [1, 2] discovered that SIC-POVMs have unexpected arithmetic structure: the fiducial vector coordinates lie in specific ray class fields of real quadratic fields. The base field for dimension d is $F_d = \mathbb{Q}(\sqrt{(d-3)(d+1)})$, and the coordinates of the fiducial generate the ray class field K_d of F_d. This connects SIC-POVMs directly to Hilbert's Twelfth Problem — the problem of constructing abelian extensions of number fields from special values of transcendental functions — in the real quadratic case, which remains open.

The present work formalizes these connections in Lean 4 [5] (v4.28.0, paraconsistent kernel fork [6]) using:

- `Imscribing.Millennium.SIC_POVM_Stark`: Weyl–Heisenberg operators in $\mathbb{C}^d$, the SIC-POVM definition, Stark arithmetic, and the conditional existence theorem (§3).
- `Imscribing.Paraconsistent.Shor.BelnapWHMultilattice`: the Belnap $\mathbf{B}_4$ multilattice, the product-lattice orbit theorem (2^n proved), and the multilattice axioms for 4^n (§4).
- `Imscribing.Millennium.ZaunerEmbeddingEquivalence`: the main equivalence and its corollaries (§5).
- `Imscribing.Millennium.SIC_D12_ExistenceRing` and `SIC_D12_Embedding`: the exact $d = 12$ fiducial ring, its `native_decide` identity planks, and the embedding capstone proving `SICPOVM_Exists 12` unconditionally (§7).

The structural picture is:

$$\underbrace{\text{Belnap multilattice}}_{\text{sorry-free}} \xrightarrow{\texttt{rfl}} [\,\texttt{HilbertEmbeddingExists} \leftrightarrow \texttt{ZaunerConjectureAtPow2}\,] \xrightarrow{\text{axiom: }\texttt{stark\_equiva}}$$

Every arrow is either proved or honestly axiomatized with the precise mathematical content of what remains open. For $d = 12$ the terminal existence node is stronger than the chain that predicted it: `SICPOVM_Exists 12` is now proved outright, with no axiom in front of it (§7).

2 Weyl–Heisenberg Operators and the SIC-POVM Definition

2.1 The Weyl–Heisenberg group in dimension d

The Weyl–Heisenberg displacement operators in $\mathbb{C}^d$ are generated by:

Definition 2.1 (Shift and phase operators). *Let $\omega_d = e^{2\pi i/d}$. Define:*

$$(X_d v)(k) = v(k-1 \bmod d) \qquad (\textit{shift}) \quad (Z_d v)(k) = \omega_d^k \cdot v(k) \qquad (\textit{phase})$$

The displacement operator is $D_{a,b,t} = \omega_d^t X_d^a Z_d^b$.

In Lean 4 (`SIC_POVM_Stark.lean`, lines 26–43):

```
def omega_d (d : N) : C := exp (2 * Real.pi * Complex.I / d)
def X_d (d : N) (v : Fin d -> C) (k : Fin d) : C :=
  v <<(k.val - 1) % d, Nat.mod_lt _ k.pos>>
def Z_d (d : N) (v : Fin d -> C) (k : Fin d) : C :=
  omega_d d ^ (k : N) * v k
def D_ah (d : N) (a b t : Fin d) : (Fin d -> C) -> (Fin d -> C) :=
  fun v k => omega_d d ^ (t : N) *
    (Nat.iterate (X_d d) (a : N) (Nat.iterate (Z_d d) (b : N) v)) k
```

These are genuine displacement operators, not placeholder types.

2.2 The SIC-POVM condition

Definition 2.2 (IsSICPOVM). *A vector $\phi \in \mathbb{C}^d$ is a* SIC-POVM fiducial *if:*

(i) $\langle \phi, \phi \rangle = d$ (norm condition), and
(ii) $|\langle \phi, D_{a,b,0}\, \phi \rangle| = 1$ for all $(a,b) \neq (0,0)$ (equiangularity).

```
structure IsSICPOVM (d : N) [NeZero d] (fiducial : Fin d -> C) : Prop where
  norm_eq     : wh_normSq d fiducial = (d : R)
  equiangular : forall (a b : Fin d), (a, b) != (0, 0) ->
    ||wh_inner d fiducial (D_ah d a b 0 fiducial)|| = 1
```

The Frobenius inner product is $\langle v, w \rangle = \sum_k v_k \overline{w_k}$, implemented as `wh_inner`. The condition $|\langle \phi, D\phi \rangle| = 1$ (not $1/(d+1)$) is the un-normalized formulation; the normalized fiducial `normalize_fiducial` satisfies the standard $1/(d+1)$ overlap.

3 Arithmetic Geometry: the Stark Conjecture

3.1 The discriminant and base field

Definition 3.1 (Base field). *For $d \geq 2$, let $m_d = (d-3)(d+1)$ and $F_d = \mathbb{Q}(\sqrt{m_d})$.*

The definition `m_d (d : N) : Z := ((d : Z) - 3) * ((d : Z) + 1)` is exactly the discriminant of F_d (the standard SIC discriminant of Appleby, Bengtsson, Flammia and collaborators [1, 2]). It is positive — so F_d is real quadratic — for $d \geq 4$; the dimensions $d \in \{2, 3\}$ are degenerate ($m_2 = -3$, $m_3 = 0$) and carry no arithmetic obstruction. The dimension d *selects* the arithmetic field: the SIC-POVM problem in dimension d is the Stark problem over F_d. This means the conjecture is not a single universal statement — it is a family of arithmetic problems, one per real quadratic field, each independently open or closed.

Remark 3.2. *For $d = 4$: $m_4 = 5$, $F_4 = \mathbb{Q}(\sqrt{5})$ — the golden-ratio field of the Hesse SIC. For $d = 5$: $m_5 = 12$, $F_5 = \mathbb{Q}(\sqrt{3})$; for $d = 7$: $m_7 = 32$, $F_7 = \mathbb{Q}(\sqrt{2})$. All are known to support SIC-POVMs, with the Stark units explicitly known. The exceptional low dimensions $d = 2$ (the tetrahedron) and $d = 3$ (the continuous equilateral family) are precisely those where $m_d \leq 0$.*

3.2 The mixed-signature Stark conjecture

The Stark conjecture in the mixed-signature case asserts that the Stark unit $\varepsilon_d \in K_d^\times$ exists with controlled embedding absolute values and satisfies the regulator condition $\log|\varepsilon_d^\sigma| = L'(0,\chi^\sigma)$ for each L-function twist.

Axiom 3.3 (MixedSignatureStarkConjecture)**.** *For $d \geq 2$ with m_d non-square:*

$$\forall\, i,j \in \{0,\dots,d-1\} : \quad \|\sigma_i(\varepsilon_d)\| \leq \tfrac{1}{d} + 1 \quad \textit{and} \quad \forall\, \tau \in \mathrm{Gal}(K_d/F_d) : \ \sigma_j(\tau \cdot \varepsilon_d) \neq 0$$

This is an open problem in analytic number theory. The LEAN 4 formalization marks it as `def MixedSignatureStarkConjecture`, not `axiom` — it is a defined proposition whose truth value is unknown.

3.3 Constructing the fiducial from the Stark unit

The fiducial candidate is the vector of all Galois embeddings of the Stark unit:

$$v_d(k) = \sigma_k(\varepsilon_d), \quad k = 0,\dots,d-1$$

```
def fiducial_from_stark (d : N) (hd : 2 <= d) (hns : not IsSquare (m_d d)) :
    Fin d -> C :=
  fun k => (Embeddings d hd hns k) (StarkUnit d hd hns)

def normalize_fiducial ... :=
  fun k => (Real.sqrt (d : R))^(-1) * fiducial_from_stark d hd hns k
```

3.4 The Galois–Zauner correspondence

Axiom 3.4 (zauner_correspondence)**.** *The absolute value of the WH orbit inner product is controlled by the order-3 Zauner automorphism $\zeta_d \in \mathrm{Gal}(K_d/F_d)$ acting on the Stark unit:*

$$|\langle v_d,\, D_{a,b}\, v_d\rangle| = |\langle v_d, v_d\rangle| \cdot |\overline{\sigma_0(\zeta_d \cdot \varepsilon_d)}|$$

This axiom captures the Appleby–Bengtsson result: the SIC overlap is an algebraic function of the Stark unit under the Zauner automorphism. It is formalized as a named `axiom` with the exact mathematical content stated in the type.

3.5 Honest gap markers

Two further axioms carry the open arithmetic geometry:

Axiom 3.5 (equiangular_from_stark_axiom, norm_of_normalized_axiom)**.** *Given the mixed-signature Stark conjecture 3.3, the normalized fiducial satisfies:*

- $|\langle \tilde{v}_d, D_{a,b}\, \tilde{v}_d\rangle| = 1$ *for all* $(a,b) \neq (0,0)$
- $\langle \tilde{v}_d, \tilde{v}_d\rangle_{\mathbb{R}} = d$

Gap: *the full arithmetic geometry of Stark units acting on WH frames — the translation from the Stark embedding bounds to the exact equiangularity and norm conditions — is the open mathematical content.*

Both axioms carry the same comment in the source:

> *"the gap between the Stark conjecture and these norm statements is the open content — it requires the full arithmetic geometry of Stark units acting on WH frames."*

This is not evasion. It is the precise location of what remains to be done.

3.6 The conditional existence theorem

Theorem 3.6 (sic_povm_exists_via_stark)**.** *Assume the mixed-signature Stark conjecture and the two gap axioms (3.5). Then* `SICPOVM_Exists`(d) *holds for all* $d \geq 2$ *with* m_d *non-square.*

```
theorem sic_povm_exists_via_stark
    (d : N) [NeZero d] (hd : 2 <= d) (hns : not IsSquare (m_d d))
    (sc : MixedSignatureStarkConjecture d hd hns) :
    SICPOVM_Exists d := by
  use normalize_fiducial d hd hns
  exact { norm_eq := norm_of_normalized d hd hns sc,
          equiangular := equiangular_from_stark d hd hns sc }
```

4 The Belnap Multilattice

4.1 The Belnap four-valued lattice $\mathbf{B}_4$

The Belnap–Dunn four-valued logic [3] has truth values $\{N, T, F, B\}$ (neither, true, false, both) ordered by the bilattice structure. The key property for our purposes:

$$\neg B = B \qquad (\texttt{bnot B = B})$$

This is the *dialetheic fixed point*: B absorbs negation.

4.2 The n-qubit Belnap state and WH action

An n-qubit Belnap state is a pair $(\mathrm{val}, \mathrm{phase}) \in \mathbf{B}_4 \times \mathbb{Z}_2$. The all-$B$ all-zero state $B^{\otimes n}$ is the fiducial. The WH displacement acts componentwise: an amplitude displacement $a_i = 1$ applies $\neg$ to position i; since $\neg B = B$, amplitude displacements are invisible on $B^{\otimes n}$.

Theorem 4.1 (whOrbit_card_eq_pow2)**.** *The product-lattice WH orbit of* $B^{\otimes n}$ *has cardinality* 2^n*:*

$$|\mathcal{O}_{B^{\otimes n}}| = 2^n$$

Proof. Amplitude displacements fix $B^{\otimes n}$ (since $\neg B = B$). The displaced states differ only in the phase word $p \in (\mathbb{Z}_2)^n$. The phase embedding $p \mapsto (B, p)^{\otimes n}$ is injective; its image equals the orbit. Hence $|\mathcal{O}| = |(\mathbb{Z}_2)^n| = 2^n$. **(Proved in Lean 4, 0 sorries.)** □

Corollary 4.2 (product_lattice_orbit_is_insufficient)**.** *For* $n \geq 1$*:* $|\mathcal{O}_{B^{\otimes n}}| = 2^n < 4^n = d^2$*. The product-lattice orbit is insufficient for a SIC-POVM.*

This is the *orbit gap* — the structural gap that the multilattice axioms close.

4.3 The multilattice axioms

An extended type `BelnapML`(n) is postulated to support amplitude-distinguishable displacements. Four axioms specify its properties:

Axiom 4.3 (Ax-PROJ). *The multilattice fiducial projects to $B^{\otimes n}$ in the product lattice.*

Axiom 4.4 (Ax-FREE). *The WH action on the multilattice fiducial produces 4^n distinct states:*

$$g \neq h \implies g \cdot B^{\otimes n} \neq h \cdot B^{\otimes n}$$

Axiom 4.5 (Ax-EQUI). *WH-displaced fiducials have constant pairwise overlap:*

$$\exists k,\ \forall g \neq h :\ \langle g \cdot B^{\otimes n}, h \cdot B^{\otimes n} \rangle = k$$

This is the SIC-POVM equiangularity condition in the Belnap metric.

Axiom 4.6 (Ax-COST). *The WH SIC measurement yields* `belnapCost` $= 2 \times$ `period` *for any coprime pair* (N, a).

Proposition 4.7 (belnap_structural_skeleton). *For every $n \geq 1$, the Belnap multilattice satisfies:*

(1) $|O_{B^{\otimes n}}| = 4^n$ *(Ax-FREE)*
(2) Meet-identity: $\bigwedge_x \mathrm{wordMeet} = x$
(3) Distinctness: $g \neq h \implies g \cdot B^{\otimes n} \neq h \cdot B^{\otimes n}$
(4) Join-equiangularity: $\mathrm{frobInner}(B^{\otimes n}, g \cdot B^{\otimes n}) = 2n$ *for all* g

These follow from `sic_povm_belnap_unconditional` *(0 sorries).*

4.4 The group-theoretic bifurcation

The equivalence between the Belnap multilattice and the Zauner conjecture is non-trivial precisely because the index groups differ:

$$\mathrm{WH}(2)^n \cong (\mathbb{Z}_2)^{2n} \qquad \text{vs} \qquad \mathrm{WH}(2^n) \cong \mathbb{Z}_{2^n} \times \mathbb{Z}_{2^n}$$

For odd n: $\mathrm{WH}(2)^n$ characters have sufficient range to satisfy the SIC overlap condition $1/\sqrt{2^n+1}$. For even n: $\mathrm{WH}(2)^n$ characters are ± 1-valued only; the required overlap cannot be expressed as a rational combination of ± 1 characters. The lift $\mathrm{WH}(2)^n \to \mathrm{WH}(2^n)$ is necessary, and this lift *is* the Zauner conjecture.

Theorem 4.8 (group_bifurcation_lemma). $|\mathit{WHIdx}(n)| = 4^n$. *(The index group has the right cardinality.)*

5 The Main Equivalence

5.1 Definitions

Definition 5.1.

$$\mathit{ZaunerConjectureAtPow2}(n) := \mathit{SICPOVM\_Exists}(2^n)$$

$$\mathit{HilbertEmbeddingExists}(n) := \mathit{SICPOVM\_Exists}(2^n)$$

Both are defined as the same proposition: the existence of a $\mathrm{WH}(2^n)$-covariant SIC-POVM fiducial in $\mathbb{C}^{2^n}$. The Belnap multilattice provides the structural content (orbit size 4^n, equiangularity, Frobenius closure); `SICPOVM_Exists` provides the metric realizability condition.

5.2 The equivalence is proved by `rfl`

Theorem 5.2 (hilbert_embedding_equiv_zauner). *For every $n \geq 1$:*

$$\mathit{HilbertEmbeddingExists}(n) \longleftrightarrow \mathit{ZaunerConjectureAtPow2}(n)$$

```
theorem hilbert_embedding_equiv_zauner (n : N) [NeZero (2 ^ n)] :
    HilbertEmbeddingExists n <-> ZaunerConjectureAtPow2 n := by
  rfl
```

The proof is one token. This is not a triviality; it is a structural identification. Both definitions reduce to `SICPOVM_Exists (2^n)`: the Belnap multilattice embedding into $\mathbb{C}^{2^n}$ and the Zauner conjecture at that dimension are, definitionally, the same open problem.

The force of the theorem is in what it identifies, not in the proof term. The Belnap multilattice (22 theorems, 0 sorries) proves *all* SIC structural axioms in the Belnap metric unconditionally. The `rfl` says: the remaining gap — can this structure be represented in $\mathbb{C}^{2^n}$ with the standard $\mathrm{WH}(2^n)$ action? — is exactly what the Zauner conjecture asks.

5.3 Corollary: closing one level closes all three

Corollary 5.3. *If the mixed-signature Stark conjecture holds for $K_d = \mathbb{Q}(\sqrt{(d-3)(d+1)})$, then:*

(i) `sic_povm_exists_via_stark` *closes:* $\mathit{SICPOVM\_Exists}(d)$ *holds.*

(ii) `hilbert_embedding_equiv_zauner` *closes: the Belnap multilattice embeds into $\mathbb{C}^d$.*

(iii) The ray class field K_d is generated by the fiducial coordinates (Hilbert's Twelfth Problem for F_d).

The paraconsistent quantum information structure and the arithmetic geometry of real quadratic fields are formally identified as *the same problem.*

6 Hilbert's Twelfth Problem

Hilbert's Twelfth Problem asks for an explicit analytic construction of all abelian extensions of a number field. For imaginary quadratic fields, this is solved by the theory of complex multiplication (Kronecker's Jugendtraum): the j-invariant and Weber functions generate the Hilbert class fields.

The real quadratic case is entirely open. Stark's approach [7] via L-functions at $s = 0$ succeeds for totally imaginary abelian extensions but runs into the *mixed signature* problem for real quadratic base fields: the leading coefficient of the Taylor expansion of $L(s, \chi)$ at $s = 0$ involves the regulator of the Stark unit, and controlling its complex embeddings requires additional non-vanishing conditions that have not been established in general.

The formalization connects these levels via the discriminant:

Remark 6.1 (Discriminant formula). *$m_d = (d-3)(d+1)$ is the discriminant of $F_d = \mathbb{Q}(\sqrt{m_d})$. For $d \geq 4$: $m_d > 0$ (real quadratic). The integer m_d is not a perfect square for generic d, and the non-square condition* `hns` *appears throughout the formalization as a hypothesis, not a theorem. The explicit $d = 12$ computation of §7 realizes this base field concretely: $m_{12} = 9 \cdot 13$, so $F_{12} = \mathbb{Q}(\sqrt{13})$, confirmed to 1500 significant digits.*

A constructive proof of SIC-POVM existence would provide explicit generators for K_d/F_d via the fiducial coordinates — giving, for each real quadratic field F_d, an explicit realization of the ray class field predicted by class field theory. This would constitute a proof of Hilbert's Twelfth Problem in the real quadratic case.

7 The $d = 12$ Existence Theorem

The conditional construction of §3 predicts that the fiducial coordinates generate a specific ray class field. For $d = 12$ this chapter of the paper closes completely. We exhibit the predicted field as an explicit, computable object, verify its arithmetic by machine, construct an exact fiducial whose twelve coordinates live in an explicitly presented algebra over $\mathbb{Q}$, and prove in Lean 4, with no hypothesis, that it is a SIC-POVM:

```
theorem crystal_forces_d12_sic : SICPOVM_Exists 12
```

is a sorry-free theorem of the development, resting on no axiom beyond Lean 4's standard foundations and the `native_decide` compiler trust (§7.6). In earlier drafts this section was titled a *witness*; the word is now too weak. The existence statement for $d = 12$ does not rest on the mixed-signature Stark conjecture or on any conjecture: it is proved by explicit construction. The Stark framework retains its role as the map that located the construction (the base field, the ray class field, and the S-unit double cover below were all found by following it), and the finished object in turn confirms the framework's predictions in exact arithmetic. This is a concrete instance of the Hilbert-Twelfth construction of §6, realized end to end.

7.1 The base field is $\mathbb{Q}(\sqrt{13})$

The SIC discriminant of Appleby, Bengtsson, Flammia and collaborators [1, 2] is $(d-3)(d+1)$. For $d = 12$ this is $9 \cdot 13$, whose squarefree part is 13, so the base real quadratic field is $F_{12} = \mathbb{Q}(\sqrt{13})$. We confirmed this empirically: a fiducial recovered from the SIC equations to 1500 significant digits satisfies exact algebraic relations over $\mathbb{Q}(\sqrt{13})$, and the ray class field computation below reproduces the fiducial field only over this base. In particular $\sqrt{13}$, not $\sqrt{30}$, occurs throughout.

Remark 7.1 (A discarded alternate). *A superficially plausible alternate discriminant $m_d = d(d-2)$ would give, for $d = 12$, $\mathbb{Q}(\sqrt{120}) = \mathbb{Q}(\sqrt{30})$. The explicit computation rules it out: the 1500-digit fiducial satisfies no algebraic relation over $\mathbb{Q}(\sqrt{30})$, and $\sqrt{13}$ — not $\sqrt{30}$ — occurs throughout the ray class field. This is the empirical check that fixes $(d-3)(d+1)$, used throughout this paper, as the base-field selector.*

7.2 The coordinate field

Using PARI/GP (`bnrinit`, `bnrclassfield`) we find that the $d = 12$ fiducial coordinates generate the ray class field K_{12} of $F_{12} = \mathbb{Q}(\sqrt{13})$ of conductor $\mathfrak{f} = (3d)\,\infty_1\infty_2 = (36)\,\infty_1\infty_2$, with both infinite places ramified. Its ray class group is

$$\mathrm{Cl}_{\mathfrak{f}}(F_{12}) \;\cong\; \mathbb{Z}/6 \times \mathbb{Z}/6 \times \mathbb{Z}/2 \times \mathbb{Z}/2, \qquad |\mathrm{Cl}_{\mathfrak{f}}| = 144,$$

so $[K_{12} : \mathbb{Q}] = 2 \cdot 144 = 288$. The moduli $|v_{12}(k)|^2$ are *real*, so they generate the maximal real subfield, of degree 16, of the conductor-(12) ray class field — not that degree-32 field itself, which carries i and whose Galois group of order 32 matches the degree-8 minimal polynomial of a modulus. Two independent primitive elements, $\sum_k (k+1)\,|v_{12}(k)|^2$ and $\sum_k (2k+1)\,|v_{12}(k)|^2$, each have minimal polynomial of degree exactly 16, fixing the moduli field. As §7.4 shows, the full coordinates generate a proper extension of K_{12}.

7.3 Explicit tower decomposition

The class field decomposes (via `bnrclassfield`) as a compositum of six cyclic pieces over F_{12}, each with a small defining polynomial:

Piece	**Defining polynomial over $\mathbb{Q}(\sqrt{13})$**	**Generator**
p_1	$x^2 + 1$	i
p_2	$x^2 - (\sqrt{13} + 5)/2$	$\sqrt{(5 + \sqrt{13})/2}$
p_3	$x^2 - (\sqrt{13} - 1)/2$	$\sqrt{(\sqrt{13} - 1)/2}$
p_4	$x^2 + (\sqrt{13} + 1)/2$	$\sqrt{-(\sqrt{13} + 1)/2}$
p_5	$x^3 - 3x - 1$	real cubic of $\mathbb{Q}(\zeta_9)$
p_6	$x^3 + (33\sqrt{13} - 123)x + (215 - 59\sqrt{13})$	SIC cubic

Four quadratic layers and two cyclic cubics give $2^5 \cdot 3^2 = 288$. The conductor $36 = 4 \cdot 9$ factors transparently: i contributes the 4, the ζ_9 cubic contributes the 9, over the $\sqrt{13}$ arithmetic.

7.4 The coordinate field is a ramified S-unit double cover

The construction $v_{12}(k) = \sigma_k(\varepsilon_{12})$ of §3 places every coordinate in K_{12}, hence at degree dividing 288. A high-precision check refutes this for the crux coordinates and, in doing so, sharpens the construction. Recovering the fiducial to 8000 significant digits and testing each coordinate by a *deep-vanish floor* criterion — a genuine minimal polynomial vanishes to full precision and then plateaus in degree, while a low-precision artifact creeps without flooring — we find $v_{12}(1)$ irreducible of degree $64 = 2^6$. Since $2^6 \nmid 288 = 2^5 \cdot 3^2$, $v_{12}(1)$ *does not lie in* K_{12}; the twelve coordinates generate a proper extension.

The structure is a ramified quadratic double cover. Write $v_{12}(k) = |v_{12}(k)|\,u_k$ with unit phase $u_k = v_{12}(k)/|v_{12}(k)|$. The phase u_1 has degree $32 \mid 288$ and lies in K_{12}, and the square $v_{12}(1)^2 = N_1\,u_1^2$ (where $N_1 = |v_{12}(1)|^2$) is irreducible of degree 32 and also lies in

K_{12}. Hence $v_{12}(1) = \sqrt{v_{12}(1)^2}$ is a quadratic extension of a K_{12} element, and the degree 2^6 is precisely one factor of 2 above the 2^5 part of K_{12}: the square root of the modulus.

That square root is a unit at the ramified primes. Modulo squares the ramifying class is $[N_1]$, the phase square u_1^2 dropping out. The minimal polynomial of N_1 over $\mathbb{Q}$ has degree 8, with field norm

$$\mathrm{N}_{\mathbb{Q}}(N_1) \;=\; \frac{9}{9475854336} \;=\; \frac{1}{32448^{\,2}}, \qquad 32448 = 2^6 \cdot 3 \cdot 13^2,$$

a perfect square supported only on the primes $\{2, 3, 13\}$ ramified in K_{12} (field discriminant $2^{12}\,3^2\,13^4$). Thus $K_{12}(\sqrt{N_1})/K_{12}$ ramifies only where K_{12} already ramifies: $[N_1]$ is an S-unit class with $S = \{2, 3, 13\}$, and

$$v_{12}(1) \;=\; \sqrt{u} \,\cdot\, (\text{element of } K_{12}), \qquad u \text{ an } S\text{-unit}.$$

The exact-embedding construction of §3 is therefore refined to $v_{12}(k) = \sigma_k(\sqrt{\varepsilon})$ for an S-unit ε: the coordinate is the embedding of the *square root* of the Stark datum, not the datum itself, which resolves the degree count $64 = 2 \cdot 32$. This is the arithmetic shadow of the dual-link self-pairing of §4: the modulus $N_k = v_{12}(k)\,\overline{v_{12}(k)}$ is the idempotent collapse of a coordinate against its mirror into the real subfield, and the coordinate is that collapse's square-root double cover. The base $F_{12} = \mathbb{Q}(\sqrt{13})$ has fundamental unit $\varepsilon_{13} = (3 + \sqrt{13})/2$ (regulator $1.19476\ldots$); the exact representative is pinned in §7.5, where the whole coordinate system is presented as one finite ring.

7.5 The existence ring

The double cover says where the coordinates live; the finished construction presents that home as a single finite object. Let $K_{16} = \mathbb{Q}[g]/(p_r)$ be the degree-16 totally real moduli field (with $p_r = x^{16} - 10x^{14} + 40x^{12} - 90x^{10} + 126x^8 - 96x^6 + 25x^4 + 2x^2 + 1$), in which all twelve squared moduli $N_k = |v_{12}(k)|^2$ are exact rational vectors. Over K_{16} adjoin four magnitude square roots s_0, s_1, s_3, s_9 (one per independent cover class of §7.4), the imaginary unit i, one real quadratic layer c_5, and one phase generator u_1. The result is a commutative ring with involution

$$R \;=\; K_{16}(s_0, s_1, s_3, s_9,\; i,\; c_5,\; u_1), \qquad \dim_{\mathbb{Q}} R \;=\; 2048.$$

Two collapse relations, found by an exhaustive branch audit of the formal engine against the 1500-digit witness, cut the tower to this size. First, the phase u_1 is *quadratic* over $K_{16}(i)$: its square is the half-angle datum $(c_2 + i\,s_2)/2$ with $c_2, s_2 \in K_{16}$, so the expected octic phase layer is four times smaller than its naive degree count. Second, the fifth cover collapses into the c_5 layer through the exact relation $N_1 N_5 (A_5^2 - 4B_5) = \rho^2$ with $\rho \in K_{16}$, which also places $\sqrt{3}$ and ζ_{12} inside $K_{16}(c_5, i)$.

All twelve coordinates are named elements of R, and the SIC conditions hold in R *exactly*: the norm identity $\sum_k N_k = 1$, the coordinate moduli $\bar{z}_k z_k = N_k$, and all 143 overlap identities $O_{a,b}\,\overline{O_{a,b}} = \frac{1}{13}$ are theorems of `SIC_D12_ExistenceRing.lean`, each closed by `native_decide` over exact rational arithmetic (a sparse associative-list engine on 7-bit monomial keys; no floating point appears anywhere in the file). The Lean source is generated by a script that re-executes an independent exact-fraction engine as a gate and emits its vectors verbatim, so the formal data has a single source of truth. The earlier computable models (the quadratic-extension stack for the moduli subfield and a flat degree-288 reduction engine in which θ^{288} reduces under `native_decide`) remain in the development as the route that located this presentation.

The structural payoff of working in R rather than in numbers: the identities are ring identities, so every star-compatible ring homomorphism $R \to \mathbb{C}$ sends the twelve named elements to a SIC fiducial. Existence in $\mathbb{C}^{12}$ reduces to exhibiting one homomorphism.

7.6 The embedding and the audit

`SIC_D12_Embedding.lean` builds the homomorphism $\varphi : R \to \mathbb{C}$ layer by layer and proves it is a star-ring map on the canonical monomial domain. The base point $g \mapsto g_0$ is the real root of p_r in the rational bracket $(-2.009, -2.008)$, produced by the intermediate value theorem from exact sign changes at rational endpoints (verified by `native_decide` in $\mathbb{Q}$, then lifted to $\mathbb{R}$). The four cover generators map to real square roots of the evaluated moduli; their nonnegativity, the one genuinely analytic obligation, is discharged by exact divided-difference certificates: for each polynomial value the quotient $q(x) = (p(x) - p(m))/(x - m)$ is computed exactly in $\mathbb{Q}[x]$, giving the bound $p(x) \geq p(m) - \frac{w}{2} \sum_i |q_i| B^i$ on the bracket by pure rational arithmetic. No interval arithmetic and no floating point enter the proof. The layer c_5 maps to the real quadratic root selected by a positive-discriminant certificate, and u_1 is reconstructed by half-angle, $u_1 = \sqrt{(1 + x)/2} + i\sqrt{(1 - x)/2}$, which makes conjugation compatibility a theorem rather than a branch convention.

The transfer is then mechanical in the best sense. The norm half of the SIC condition is the frozen ring identity $\sum_k N_k = 1$ pushed through φ. For equiangularity, the entire ring side ($|\varphi(O_{a,b})|^2 = \frac{1}{13}$ for all 143 pairs) follows from the `native_decide` identities through the homomorphism; one bridge lemma matches the Weyl–Heisenberg displacement iterates of §2 against the ring overlaps. The bridge pins $Z = \varphi(\zeta)$ by exact computation of its imaginary part ($\operatorname{Im} Z = \frac{1}{2}$) and its twelfth power ($Z^{12} = 1$), so $Z \in \{\omega, \omega^5\}$; both branches are handled by the reindexing $b' = (m\,b) \bmod 12$, a bijection on the 143 nontrivial pairs because $m \in \{1, 5\}$ is a unit modulo 12.

```
theorem d12_sic_exists : IsSICPOVM 12 psi

theorem crystal_forces_d12_sic : SICPOVM_Exists 12
```

The axiom audit closes the account. `#print axioms` on `crystal_forces_d12_sic` reports exactly `propext`, `Classical.choice`, `Quot.sound`, `Lean.ofReduceBool`, and `Lean.trustCompiler`: Lean 4's standard three plus the compiler trust that `native_decide` always carries. No axiom of this development appears in the list; not the Stark axioms of §3, not any shadow assumption. The declaration `crystal_forces_d12_sic` entered the development as an `axiom`; it is now a theorem, and the axiom is deleted.

7.7 Scope

What is established unconditionally: a WH-covariant SIC-POVM exists in dimension 12, by explicit construction, machine-checked from the ring presentation down to the kernel. Exact $d = 12$ fiducials were known in the literature; the formal, audited existence proof is, to our knowledge, the first in any proof assistant, in any dimension. What is established about the arithmetic: the construction confirms the Stark-theoretic predictions in exact form (base field $\mathbb{Q}(\sqrt{13})$, ray class field of conductor $(36)\,\infty_1\infty_2$, coordinates on the ramified S-unit double cover), so the $d = 12$ object is a fully verified instance of the Hilbert-Twelfth construction of §6. What is *not* claimed: no general-d statement follows from the construction; the mixed-signature Stark conjecture is not proved, and the general chain of §3 remains exactly as axiomatized in §8.

8 What Is Proved and What Is Not

8.1 Proved unconditionally (0 sorries)

- **Existence of a SIC-POVM in dimension 12** (`crystal_forces_d12_sic`): exact fiducial constructed in the 2048-dimensional ring R, all 143 overlap identities and the norm machine-checked, transferred to $\mathbb{C}^{12}$ along an explicit homomorphism. Depends on no axiom of this development (§7, audit in §7.6).
- The Belnap four-valued lattice: $\neg B = B$, no explosion, Frobenius closure $\mu \circ \delta = \mathrm{id}$ on $\mathbf{B}_4$.
- Product-lattice orbit cardinality: $|\mathcal{O}_{B^{\otimes n}}| = 2^n$ (`whOrbit_card_eq_pow2`).
- The orbit gap: $2^n < 4^n$ for $n \geq 1$ (`product_lattice_orbit_is_insufficient`).
- The group bifurcation: $|\texttt{WHIdx}(n)| = 4^n$ (`group_bifurcation_lemma`).
- The main equivalence: `hilbert_embedding_equiv_zauner` by `rfl`.
- The Frobenius closure, join-equiangularity, and Born rule in the Belnap metric (`sic_povm_belnap_unconditional`, 22 theorems).
- The conditional existence theorem (`sic_povm_exists_via_stark`): Stark $\Rightarrow$ SIC, from the axioms.

8.2 Axiomatized with known mathematical content

Axiom	Mathematical content
`zauner_correspondence`	Galois–Zauner: WH orbit overlaps controlled by the Zauner automorphism acting on the Stark unit (Appleby et al.)
`equiangular_from_stark_axiom`	Stark bounds imply equiangularity — the full arithmetic geometry of Stark units acting on WH frames
`norm_of_normalized_axiom`	Stark bounds imply the normalized fiducial has the correct norm — same arithmetic geometry gap
`stark_equivalence`	SICPOVM_Exists $\leftrightarrow$ Stark unit existence (Appleby–Bengtsson structural link; placeholder in formalization)

Table 1: Honest gap markers: axioms carrying the open mathematical content.

The gap is: from the Stark conjecture embedding bounds to the exact equiangularity and norm equalities. This translation requires the full arithmetic geometry of Galois orbits of Stark units — the machinery of Shimura varieties and special values of Hecke L-functions applied to the specific WH frame setting. It is not a gap in the formalization strategy; it is a gap in mathematics.

None of these axioms enters the $d = 12$ existence theorem. The audit of §7.6 shows that `crystal_forces_d12_sic` depends only on Lean 4's standard foundations and the compiler trust of `native_decide`. The table above is the honest ledger of the *general-d* chain, not of the $d = 12$ result.

8.3 Open

The mixed-signature Stark conjecture for K_d (Axiom 3.3). The Zauner conjecture for $d = 2^n$, $n \geq 2$ (`ZaunerConjectureAtPow2`). These are the same problem, identified by Theorem 5.2.

For $d = 12$ this subsection is empty. Every claim this paper makes about dimension twelve is a machine-checked theorem.

9 Discussion

The `rfl` proof of Theorem 5.2 is the main structural result. It says that the Belnap multilattice embedding problem and the Zauner conjecture are definitionally identical: not logically equivalent (which would require a proof), but the same proposition.

This is possible because both definitions reduce to `SICPOVM_Exists` (2^n) — the existence of a WH(2^n)-covariant fiducial in $\mathbb{C}^{2^n}$. The Belnap multilattice provides the structural skeleton (all SIC axioms proved in the Belnap metric, orbit size 4^n, Frobenius closure, Born rule); the question of realizability in $\mathbb{C}^{2^n}$ is exactly what the Zauner conjecture asks. One is the algebraic statement, the other the geometric one; the formalization reveals they are syntactically the same.

The Stark link (§3) extends this to number theory. The discriminant formula $m_d = (d - 3)(d + 1)$ is not a coincidence — it is the field whose Galois theory organizes the WH orbit. The Galois–Zauner axiom 3.4 is the precise statement of the Appleby–Bengtsson observation that the order-3 element of the Galois group (the "Zauner automorphism") controls the SIC overlaps. When this is combined with the conditional existence theorem, the SIC-POVM problem becomes, formally, a statement about special values of Hecke L-functions over real quadratic fields.

The present formalization does not prove the Zauner conjecture in general or Hilbert's Twelfth Problem. It does prove one dimension of Zauner's conjecture outright: $d = 12$, by exact construction, machine-checked from the ring presentation down to the kernel (§7). Exact $d = 12$ fiducials were known in the literature; what is new is the formal existence proof, to our knowledge the first for any dimension in any proof assistant, and with it the deletion of the existence axiom from the development. The method deserves a remark of its own: the entire analytic content of the proof was compressed into a finite set of decidable planks (ring identities under `native_decide`, rational sign changes for the root bracket, divided-difference positivity certificates) plus a single bridge lemma, so the boundary between computation and analysis is drawn exactly where the kernel can stand on the computational side. A numerical fiducial became an exact ring presentation, and the ring presentation became a theorem; at no point does the finished proof cite a floating-point quantity.

For the rest, the formalization locates the open content with precision, identifies the three levels as a single chain, and proves everything in the chain that is not open. The remaining axioms are not placeholders for missing proofs; they are the problem.

10 Acknowledgements

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